Back **Introduction to number bases – conversion to decimal**
Practice Assignment • 30 min

Due Jul 3, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. Why do humans typically use base 10? 1 / 1 point

- Because we have 10 toes
- Because it is the most efficient base
- Because we have five fingers on each hand
- Because we have 10 digits (fingers)

Nice work

Correct! The use of base 10 is because humans typically have 10 fingers.

2. What does the number 327 represent in base 10? 1 / 1 point

- 3 thousands, 2 hundreds and 7 tens
- 3 hundreds 2 tens and 7 ones
- 3 tens, 2 hundreds and 7 ones
- 3 ones, 2 tens and 7 hundreds

Nice work

Correct! 327 means 3 hundreds, 2 tens and 7 ones.

3. What is the base used by computers to represent numbers? 1 / 1 point

- Base 8
- Base 10
- Base 2
- Base 16

Nice work

Correct! Computers use base 2 (binary).

4. Why do computers use base 2? 1 / 1 point

- It is the most familiar to humans
- It is the most efficient base for calculations
- It was chosen arbitrarily
- It is simpler to implement with switches that have two states: on and off

Nice work

Correct! Computers use base 2 because it aligns with the binary state of electronic switches.

5. What is the general rule for any number to the power of 1? 1 / 1 point

- It equals the number itself
- It equals 1
- It equals 0
- It equals the base

Nice work

6. How is the sound wave represented digitally on a computer? 1 / 1 point

- By sampling the wave and converting it into binary numbers
- By recording it directly as a continuous signal
- Using base 8
- Using base 10

Nice work

7. In base 2, what does the binary number 1011 represent in decimal? 1 / 1 point

11

Feedback

Correct answer: 11

To convert the binary number 1011 to decimal, we need to understand the place values in base 2. Each digit in the binary number represents an increasing power of 2 starting from the rightmost digit (which represents 2^0).

Step 1: Write down the binary numbers and identify the place values:

1	0	1	1
2^3	2^2	2^1	2^0

Step 2: Calculate the value of each place:

$1 \cdot 2^3 = 1 \cdot 8 = 8$

$0 \cdot 2^2 = 0 \cdot 4 = 0$

$1 \cdot 2^1 = 1 \cdot 2 = 2$

$1 \cdot 2^0 = 1 \cdot 1 = 1$

Step 3: Sum the values:

8 + 0 + 2 + 1 = 11

Therefore, the binary number 1011 represents the decimal number 11.

8. How would you represent the decimal number 57 in binary? 1 / 1 point

111001

Feedback

Correct answer: 111001

To convert the decimal number 57 to binary, we need to repeatedly divide the number by 2 and record the remainders. Here are the steps to convert 57 to binary:

Step 1: Divide 57 by 2:

Quotient: 28

Remainder: 1 (this is the least significant bit)

Step 2: Divide the quotient (28) by 2:

Quotient: 14

Remainder: 0

Step 3: Divide the quotient (14) by 2:

Quotient: 7

Remainder: 0

Step 4: Divide the quotient (7) by 2:

Quotient: 3

Remainder: 1

Step 5: Divide the quotient (3) by 2:

Quotient: 1

Remainder: 1

Step 6: Divide the quotient (1) by 2:

Quotient: 0

Remainder: 1 (this is the most significant bit)

Now, write the remainders down in reverse order (from the most significant bit to the least significant bit):

$57_{10} = 111001_2$

So, the binary representation of 57 is 111001.

1 / 1 point

9. Convert the decimal number 45 to binary.

101101

Feedback

Correct answer: 101101

To convert the decimal number 45 to binary, we need to repeatedly divide the number by 2 and record the remainders.

$$
\begin{aligned}
&45 \div 2 = 22 \text{ remainder } 1 \\
&22 \div 2 = 11 \text{ remainder } 0 \\
&11 \div 2 = 5 \text{ remainder } 1 \\
&5 \div 2 = 2 \text{ remainder } 1 \\
&2 \div 2 = 1 \text{ remainder } 0 \\
&1 \div 2 = 0 \text{ remainder } 1
\end{aligned}
$$

Now, write the remainders down in reverse order: 101101

Therefore, 45 in binary is 101101.

10. Convert the binary number 100111 to its decimal equivalent. Show your workings. 1 / 1 point

39

Feedback

Correct answer: 39

Write down the binary numbers and identify the place values:

1	0	0	1	1	1
2^5	2^4	2^3	2^2	2^1	2^0

Step 1: Calculate the value of each place:

$1 \cdot 2^5 = 1 \cdot 32 = 32$

$0 \cdot 2^4 = 0 \cdot 16 = 0$

$0 \cdot 2^3 = 0 \cdot 8 = 0$

$1 \cdot 2^2 = 1 \cdot 4 = 4$

$1 \cdot 2^1 = 1 \cdot 2 = 2$

$1 \cdot 2^0 = 1 \cdot 1 = 1$

Step 2: Sum the values:

32 + 0 + 0 + 4 + 2 + 1 = 39

Therefore, 100111 in binary is 39 in decimal.

11. Convert the binary number 11010 to its decimal equivalent. Show your workings. 1 / 1 point

26

Feedback

Correct answer: 26

Write down the binary numbers and identify the place values:

1	1	0	1	0
2^4	2^3	2^2	2^1	2^0

Step 1: Calculate the value of each place:

$1 \cdot 2^4 = 1 \cdot 16 = 16$

$1 \cdot 2^3 = 1 \cdot 8 = 8$

$0 \cdot 2^2 = 0 \cdot 4 = 0$

$1 \cdot 2^1 = 1 \cdot 2 = 2$

$0 \cdot 2^0 = 0 \cdot 1 = 0$

Step 2: Sum the values:

16 + 8 + 0 + 2 + 0 = 26

Therefore, 11010 in binary is 26 in decimal.